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Carbon dioxide capture, transport and storage supply chains: Optimal economic and environmental performance of infrastructure rollout

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ABSTRACT

This work presents a novel optimization framework for the optimal design of carbon capture, transport, and storage supply chains in terms of installation, sizing and operation of carbon dioxide (CO_2) capture and transport technologies. The optimal design problem is formulated as a mixed-integer linear program that minimizes the total costs of the supply chains while complying with different emissions reduction pathways over a deployment time horizon of 25 years. All design decisions are time-dependent and are taken with a yearly resolution. Whereas the model is general, here its features are illustrated by designing optimal supply chains to decarbonize the Swiss waste-to-energy sector, for various emission reduction pathways, when up to two storage sites are considered, namely one in the North Sea assumed to be already available and a hypothetical one in Switzerland assumed to be possibly available in the future. Findings show that, unless a domestic storage site becomes available soon, the transport cost is the greatest contribution to the overall costs, followed by the capture cost, while the storage cost plays only a minor role. Pipelines are the most cost-effective mode of transport for large volumes of transported CO_2, especially when considering multi-year time horizons for the planning of the supply chains. Ship and barge connections are competitive with pipeline connections, whereas rail and truck connections are cost-optimal only when considering shortsighted time horizons or small volumes of CO_2 transported.

1. Introduction and scope

Limiting global warming to 1.5 °C requires a drastic reduction of anthropogenic greenhouse gas (GHG) emissions (mostly carbon dioxide, CO_2), to reach net-zero CO_2 emissions by 2050 IPCC (2018, 2021). Carbon capture and storage (CCS) can help achieving this ambitious target by limiting fossil-based emissions from hard-to-abate sectors (i.e., cement, iron and steel, chemical production, and waste treatment), as well as by removing CO_2 from the atmosphere when combined with direct air capture and bioenergy. Early CCS projects adopted a point-to-point model, consisting of a single large emitter (e.g., a fossil-based power plant) located at a reasonable distance from a CO_2 storage site (NRG, 2021; SaskPower, 2021). However, most recently, CCS hubs and clusters have been developed to aggregate CO_2 streams captured at different facilities for conditioning, transport and storage. Examples of such hubs in Europe are those being developed around the North Sea (e.g., in Norway, in the Netherlands, and in the UK) and in Iceland (Carbfix, 2021; Equinor, 2019; Porthos Project, 2021). These hubs will enable significant reductions in unit costs of CO_2 transport and storage through economies of scale, thus limiting investment risks and unlocking opportunities for CCS implementation at small CO_2 sources (Global CCS Institute, 2020). With point-source CO_2 emitters from various sectors being spread all over Europe and offering the possibility of capturing ca. 1.4 $GtCO_2/y$ (Endrava, 2020), a pan-European shared CO_2 network infrastructure connecting emission sources and storage hubs will be key to allow *returning* these CO_2 emissions to the subsurface in an effective and timely manner. Therefore, the optimal design and development of carbon capture, transport and storage (CCTS) supply chains and networks is a topic of the utmost urgency and practical relevance, and it has been investigated in a number of studies. In the following, we opt to use the term CCTS in contrast to the conventionally used term CCS to reflect the key importance of CO_2 transport for the successful implementation of such supply chains.

Most studies focused on the design of cost-optimal single CO_2 supply

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chains or networks, which meet specified environmental goals (Bakken and Velken, 2008; Bjerketvedt et al., 2020; d'Amore and Bezzo, 2017; Elahi et al., 2014; Jakobsen et al., 2014; Middleton et al., 2020; Morbee et al., 2012; Oei et al., 2014; Wei et al., 2021). For example, Wei et al. proposed a layout for a global CCS supply chain in line with a 2°C climate target, which is based on the assessment of potential emissions and storage facilities and on their vicinity (Wei et al., 2021). Detailed analyses, based on mathematical optimization, for the techno-economic assessment of CCS supply chains have also been proposed (d'Amore and Bezzo, 2017; Jakobsen et al., 2014; Middleton et al., 2020). As an example, D'Amore et al. presented a mathematical framework for the economic optimization of CCS supply chains with a European scope (d'Amore and Bezzo, 2017), and Middleton et al. developed an open-source tool for optimizing a CCS infrastructure (Middleton et al., 2020). Nevertheless, most of these studies focused on the design and assessment of optimal CO_2networks rather than on their optimal evolution over time. Those works that considered the time-dependent development of CO_2 supply chains and networks, focused solely on CO_2 transport via pipelines and/or ships (d'Amore and Bezzo, 2017; Elahi et al., 2014; Morbee et al., 2012; Oei et al., 2014). Furthermore, the analysis of the relationship of optimal CO_2 network choices, volume of CO_2 transported, and time-dependent constraints on system carbon emissions is missing.

We aim at filling this gap by (i) developing an optimization model for the cost-optimal and time-resolved design of multi-modal CCTS supply chains complying with specified emissions reduction targets, and (ii) expanding the current knowledge on CCTS supply chains by unveiling the rationale behind optimal design choices. Furthermore, we aim at improving the state-of-the-art of optimal CCTS supply chains design by designing and assessing cost-optimal CCTS supply chains that are resilient to failures. While the present work focuses on the former objectives, ongoing research is being carried out to address the latter.

The optimization problem is formulated as a mixed integer linear program (MILP), and builds upon the work of (Gabrielli et al., 2020; 2018a; 2018b). The model is flexible, general and could be applied to a multitude of industrial sectors and geographic locations. For the sake of simplicity but without loss of generality, it is illustrated here by investigating a specific case, i.e., the optimal design of a CO_2 supply chain to decarbonize the Swiss waste-to-energy (WtE) sector. It is worth noting that a key novelty of this work is the application of the model and of the optimization algorithm for assessing the dynamic development, i.e., the roll out, of such supply chain over a time horizon of 25 years, thus tackling some of the key challenges at the interface between climate policy and infrastructure development.

Investigating CCTS for the Swiss WtE sector appears interesting and worth exploring from different perspectives. First, Switzerland aims to cut its greenhouse gas emissions to net zero by 2050, thus making a contribution to the Paris Agreement consistent with its climate policy responsibilities and capabilities. The net-zero target requires the use of CCTS technologies, particularly at large, fixed-location point sources, such as WtE facilities and cement factories. However, due to Switzerland's uncertain prospects as regards CO_2 geological storage (there is no proven capacity now, and developing national CO_2 storage capacity for large-scale use will realistically require not less than 15 to 20 years), the successful implementation of CCTS will - at least in the near future - be dependent on access to storage sites abroad. The geographical features of Switzerland, a mountainous and landlocked country located in Central Europe, and its relative remoteness from soon available CO_2 storage hubs make the design of a Swiss CO_2 supply chain challenging but potentially highly insightful. In addition, the Swiss WtE sector, which comprises of a total of thirty plants located across Switzerland and emitting ca. 4.5 $MtCO_2$/year, is already focusing intensively on CCTS with a view to carbon-neutral operation and is actively promoting the launch of the first concrete projects (KVA Linth, 2022). Finally, the fraction of biogenic carbon relative to the total carbon content of the waste, which is ca. 52%, makes CCTS combined with WtE plants a promising negative emission technology (NET) solution that can help balance the residual hard-to-abate emissions caused by other sectors (VBSA, 2019; Verband der Betreiber Schweizerischer Abfallverwertungsanlagen (VBSA), 2021).

The article is structured as follows. The system configuration is described in Section 2, whereas the formulation of the multi-objective optimization problem is presented in Section 3. The main results are presented in Sections 4 and 5, for the static and dynamic cases, respectively. Finally, conclusions are drawn in Section 6.

2. System description

The CCTS supply chains developed and analyzed in this work aim at decarbonizing the Swiss WtE sector, while delivering net-negative CO_2 emissions. The CO_2 is captured at the industrial sites, i.e., WtE plants, and is transported via a multi-modal network (i.e., based on several possible modes of transportation) to the storage sites, where it is permanently sequestered underground. The CCTS supply chains are represented as a network, where the nodes consist of the CO_2 capture sites, CO_2 storage sites and transport exchange sites, as illustrated in Fig. 1.

CO_2 capture sites. The thirty Swiss WtE plants emit a total of ca. 4.5 $MtCO_2$/year, about half of which are generated from biogenic sources, thus offering an opportunity to deliver negative emissions through CCTS (VBSA, 2019; Verband der Betreiber Schweizerischer Abfallverwertungsanlagen (VBSA), 2021). We consider a point-source capture technology, which consists of the standard amine-based absorption process using monoethanolamine (MEA) as solvent, with a CO_2 recovery rate up to 90% (Haaf et al., 2020; Lausselet et al., 2017). Whereas determining the optimal value of the recovery rate would require the combined optimization of the CCTS supply chain and of the point-source capture unit installed at the WtE plants, as well as a broader inter-sectoral analysis, we select a 90% recovery rate based on the values achieved at large-scale industrial plants (e.g. Boundary Dam SaskPower, 2021). Furthermore, while we assume that the heat required for CO_2 capture is supplied internally by the WtE plant, we consider an electric energy requirement of 25 kWh/tCO_2 (Ahn et al., 2013).

The cost of carbon capture is generally estimated to be between 40 and 80 USD/tCO_2 for the power generation sector (IEA, 2019; Rubin et al., 2015). A representative value of 50 EUR/tCO_2 is used here. After capture, the CO_2 is conditioned based on the specific transport requirements.

CO_2 storage sites. Two CO_2 storage sites are considered, namely the Northern Lights CO_2 storage hub currently being developed in the North Sea (Equinor, 2019), and a hypothetical storage site in Switzerland. Although no CO_2 geological storage site is currently being built in Switzerland, intense research activity has focused recently on identifying suitable geological structures for permanent CO_2 sequestration (ACT, 2020; Wenning et al., 2020; Zappone et al., 2021). In the scope of this study, a specific location for the hypothetical Swiss CO_2 storage site has been selected, as shown in Fig. 1. The choice was made arbitrarily without referencing any current or future potential developments. Such arbitrary choice is nevertheless not critical here because the exact position of the potential Swiss storage site has a minor impact on the structure of the optimal CCTS supply chains, since it is anyway much closer to the emission sources than the Northern Lights storage site.

The Northern Lights storage site is assumed to be always available with no constraint on its storage capacity. In contrast, several scenarios are assessed for the prospective future availability of the Swiss CO_2 storage site, see Section 5.

Transport exchange sites. Two additional sites in Rotterdam and in Basel, with neither capture nor storage capabilities, are included to function as transit terminals. The Rotterdam harbor, which is at the center of a large project to turn it into an open-access CO_2 transport hub by 2023 (Porthos Project, 2021), connects continental Europe to the Northern Lights CO_2 storage hub. The site will be equipped with a

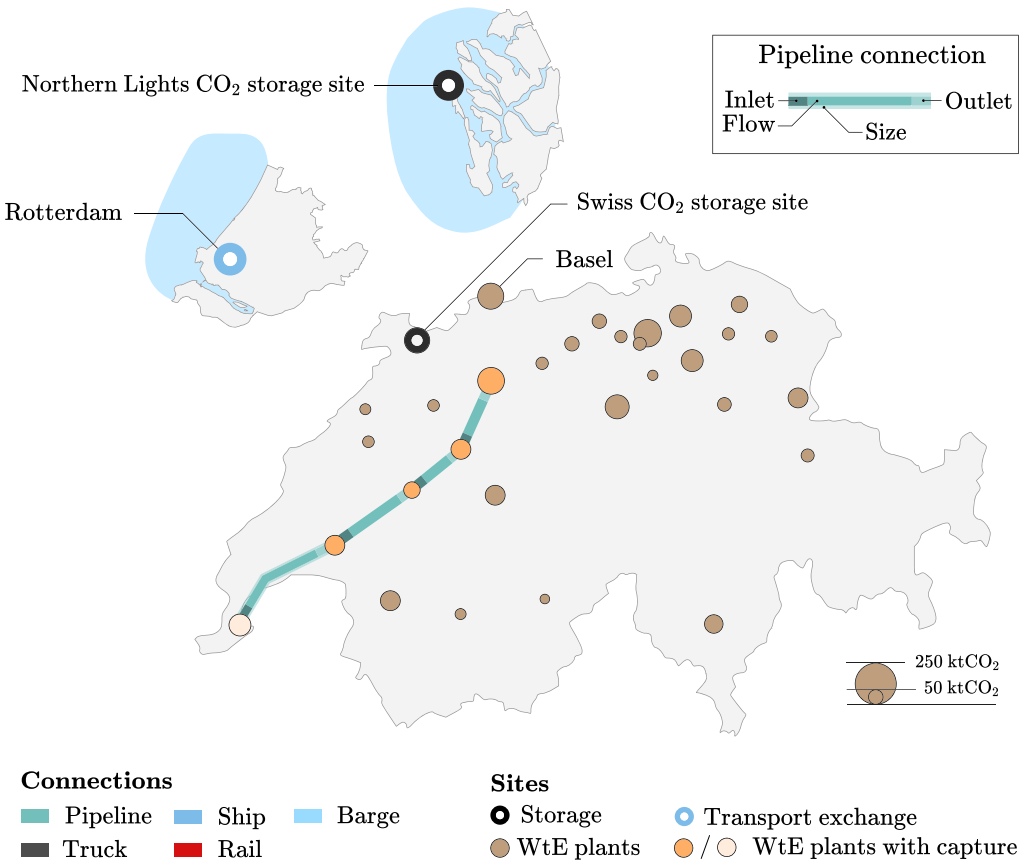


Fig. 1. Scheme of the system configuration showing: the WtE plants in Switzerland, the CO_2 storage sites (i.e., the Northern Lights site and the hypothetical site in Switzerland), the transport exchange sites (i.e., Rotterdam and Basel; note that for the latter the transport exchange site is combined with the WtE plant into a single node), few illustrative pipeline connections. For illustrative purposes, some WtE plants are represented as equipped with a CO_2 capture unit, with the light-shaded yellow dot indicating that the CO_2 recovery rate is smaller than 90%. The inset on the top right exemplifies the representation of a pipeline connection with the light-shaded area indicating the size of the installed connection (i.e., maximum capacity installed), in contrast to the actual CO_2 flow that is transported (i.e., deployed capacity); the dark-shaded and the light-shaded ends indicate the CO_2 inlet to and outlet from, respectively, the pipeline. All connections are represented in this fashion, with line thickness proportional to the CO_2 flow and/or size of the connection. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

compressor station for the conditioning of incoming CO_2 for pipeline or ship transport. The transport exchange site located in Basel, which is combined with the WtE plant into a single node, serves as collection point for the Swiss CO_2 and is directly linked to the Rotterdam exchange site. We assume that this site is setup to handle the transportation switch between truck, rail, barge and pipeline.

Equipment needed for CO_2 conditioning (e.g., compressors and heat exchangers) is assumed to be available at each node (i.e., capture, storage, and transport exchange sites) to provide any work required to achieve the transport or storage specifications (i.e., including the work needed to re-condition the CO_2 upon changing the transport mode at a given node).

Multi-modal transport. All nodes can be connected via multiple modes of transport:

- Truck connections are available between sites located in Switzerland (WtE plants and Swiss storage site) and between Basel, Rotterdam and the onshore terminal of the Northern Lights storage site. The CO_2 is transported in liquid form at a pressure of 22 bar and at a temperature of -35 °C using an ISO-tank container with unit capacity of 20 tCO_2, see e.g., COMTECSWISS (2021).
- A rail connection is included between Basel and Rotterdam. In addition, certain WtE plants are connected to Basel via railway lines departing from their own private stations located on site, which allow them to load CO_2 directly onto the container attached to a train. As for truck connections, the CO_2 is transported in liquid form using ISO-tanks (at the same pressure and temperature as for truck-based transport), which can be loaded directly on rail cars.
- Greenfield pipelines can be installed between capture sites (i.e., at WtE plants), exchange transport sites (i.e., Rotterdam and Basel), and the storage sites (i.e., the Northern Lights and Swiss storage sites). The CO_2 is transported in dense phase at 110 bar and ambient temperature. To account for the complexity and the duration of pipelines construction, we allow the size (i.e., diameter) of pipelines to be expanded only after at least ten years from their installation for a given configuration. Realistically, this translates into the constraint that, if installed, a pipeline cannot be dismantled or replaced with a smaller one during the considered time horizon. For the sake of simplicity, the length of each pipeline connection corresponds to the shortest distances between two sites. Although CO_2 pipelines are not new, this transport type requires attention to be paid to design factors, overpressure protection, and leak detection, especially when occurring through populated areas (Metz et al., 2005). Furthermore, currently, there are no regulatory standards in Europe addressing the transport of CO_2 in dense or supercritical phases (HSE, 2922). Therefore, such regulatory gaps should be filled in the near future to enable the implementation of these types of CO_2 transport.
- Barge and ship connections are available between Basel and Rotterdam (along the Rhine river), and between Rotterdam and the offshore terminal of the Northern Lights storage site (located nearby the city of Bergen), respectively. Although ships specifically designed to transport CO_2 in liquid form for CCTS applications are at an early stage of development (Al Baroudi et al., 2021; Aspelund et al., 2006; Kler et al., 2016), we assume that ISO-tank containers are loaded onto conventional cargo ships (Bjerketvedt et al., 2020). As for truck- and rail-based transport, CO_2 is transported in liquid form at a pressure of 22 bar and at a temperature of -35 °C. Barge and ship share similar costs and emission values and mainly differ in terms of their average travel speed.

The parameters describing the costs and performance (technical and environmental) of all transport technologies are reported in Appendix B at the end of the paper.

3. Formulation of the optimization model

The optimal design of CCTS supply chains, in terms of time-dependent installation, sizing and operation of the CO_2 capture and transport technologies, is determined via an optimization problem that minimizes the total costs while complying with fixed CO_2 emissions targets (see Section 3.2). The optimization algorithm is formulated as a MILP and builds upon the framework previously developed by Gabrielli et al. (Gabrielli et al., 2020; 2018a; 2018b). In its general form, the MILP can be written as follows:

$$\begin{aligned} &\min_{\boldsymbol{x},\boldsymbol{y}} (\boldsymbol{c}^{'}\boldsymbol{x} + \boldsymbol{d}^{'}\boldsymbol{y}) \\ &\text{subject to} \\ &\mathbf{A}\boldsymbol{x} + \mathbf{B}\boldsymbol{y} = \boldsymbol{b} \\ &\boldsymbol{x} \geq \mathbf{0} \in \mathbb{R}^{N}, \boldsymbol{y} \in \{0,1\}^{M} \end{aligned} \tag{1}$$

where $\boldsymbol{c}$ and $\boldsymbol{d}$ are vectors that express the objective functions in terms of continuous, $\boldsymbol{x}$, and binary, $\boldsymbol{y}$, decision variables, respectively. $\mathbf{A}$ and $\mathbf{B}$ are the corresponding constraint matrices and vector $\boldsymbol{b}$ represents the constant term of the constraints. N and M are the dimensions of the decision variables $\boldsymbol{x}$ and $\boldsymbol{y}$. Binary variables $\boldsymbol{y}$ are necessary to model the nonlinear costs of capture and transport technologies, as well as their status, i.e., installed vs. not-installed. The system is geographically and temporally resolved with a yearly resolution. All aspects of the optimization problem, namely input data, decision variables, constraints, and objective function are described in detail in the Supplementary Information.

Input data. The input data to the optimization problem consists of: (i) location and current CO_2 emissions of WtE plants, (ii) location and capacity of storage sites, (iii) performance, carbon footprint, and costs of capture, storage and transport technologies, (iv) availability of transport technologies (i.e., connectivity between nodes), (v) price and region-specific carbon intensity of electricity, and (vi) target CO_2 emissions for the overall WtE sector (thus reflecting the climate policy strategy). Detailed information on the input data (e.g., costs and emissions functions and parameters, sources, etc.) are provided in the Supplementary Information.

Decision variables. Based on the input data above, the optimization problem determines the (i) selection, location and size of CO_2 capture and transport technologies, (ii) input and output electricity and CO_2 streams of production technologies, (iii) CO_2 flow for installed transport technologies, (iv) energy expenditure for CO_2 transport.

Constraints. The constraints of the optimization problem include (i) energy and mass balances, and (ii) performance behaviour and operating limits of the capture and transport technologies.

Objective function. The optimization problem minimizes the total costs of the system while complying with specific CO_2 emissions targets for the whole system. The total costs consist of annualized investment costs, operation costs, and maintenance costs (see Supplementary Information). The resulting multi-objective optimization problem is solved via the ϵ-constraint method proposed by Mavrotas (Mavrotas, 2009).

3.1. Economic indicators

The economic performance of the CCTS supply chains is assessed by referring to two cost indicators. Both of them rely on the present value of the cost of the CCTS infrastructure in year τ, J_τ, which is expressed as:

$$J_\tau = \sum_{t=0}^{\tau} \frac{C_t + O_t + M_t}{(1+r)^t} \tag{2}$$

where C_t, O_t and M_t are the capital, operation and maintenance cost of the CO_2supply chain; r is the discount rate; $\tau \leq T$, which is the total number of years considered for the analysis.

The first indicator adopted is the levelized costs of stored carbon ($LCSC_\tau$) (Rubin et al., 2013). This is obtained by dividing the present cost, J_τ, by the present value of the amount of CO_2 stored, U_τ:

$$LCSC_\tau = \frac{J_\tau}{\sum_{t=0}^{\tau} \frac{U_t}{(1+r)^t}} \tag{3}$$

where U_t is the amount of CO_2 stored during year t.

The second economic indicator is the levelized cost of avoided carbon ($LCAC_\tau$), which is obtained by dividing the present cost, J_τ, by the avoided CO_2 (Roussanaly et al., 2021):

$$LCAC_\tau = \frac{J_\tau}{\sum_{t=0}^{\tau} \frac{E_t^{BAU} - E_t^{CCS}}{(1+r)^t}} \tag{4}$$

where the avoided $CO_2$is expressed as the difference between the emissions during year t without (business-as-usual, BAU) and with the CO_2infrastructure, E_t^{BAU} and E_t^{CCS}, respectively.

While the LCSC refers to a quantity, the amount of CO_2 stored, that is easy to measure directly, it does not consider the CO_2 emissions of the supply chain. In contrast, the LCAC considers all embodied emissions across the whole supply chain. The LCSC is lower than or equal to the LCAC.

3.2. Scenarios

We assess the optimal design of CCTS supply chains for two cases: (i) a static case, i.e., $T = 0$ (see Section 4), and (ii) a dynamic case considering a 25-year time horizon, i.e., $t \in \{0, 1, 2, \ldots, T\}$ with $T = 25$ (see Section 5).

The static case (i) provides a set of CCTS supply chain designs, where each design complies with a different CO_2 emissions target ranging from no target (minimum-cost, BAU design) to minimum emissions. In other words, this analysis results in a set of designs lying on a cost-emissions Pareto Front, which illustrates the trade-offs between achieving lower emissions targets and bearing higher costs. In this case, the temporal evolution of the supply chains is not considered, and each design represents an hypothetical static configuration, where costs are borne immediately. A single CO_2 storage site, i.e., the Northern Lights site, is assumed available. Since the economic indicators used to characterize the CCTS supply chains for this static case are calculated through Eqs. (3), (4) with $T = t = 0$, they are reduced to costs of stored and avoided carbon, CSC and CAC, respectively.

The dynamic case (ii) provides insights on the optimal planning and development of CCTS supply chains by assessing their cost-optimal time evolution, while satisfying given emissions reduction targets. Two scenarios characterized by different emissions reduction pathways are considered (each resulting in a different time-resolved CCTS supply chain design): (i) a *linear emissions reduction (LER) pathway* over the time horizon, where the total system emissions decrease linearly from the current value to the minimum value attainable in year T; and (ii) a *cumulative emissions reduction (CER) pathway* over the time horizon, where the cumulative total system emissions over the entire horizon are imposed to be the same as in case (i) however they are distributed in time, with the further constraint that minimum emissions are attained in year T.

The yearly values of total system emissions are imposed as a constraint for the LER pathway, while they are determined as part of the solution of the optimization problem for the CER pathway (while complying with the cumulative emissions reduction pathway). For both pathways, the Northern Lights CO_2 storage site is assumed operational over the entire time horizon, while the Swiss CO_2 storage site becomes operational only after year 15. The effect of the time when the Swiss CO_2 storage site becomes available on the structure and performance of the network is investigated through a sensitivity analysis (see Section 5.3).

4. Static design of CCTS supply chains

Fig. 2 shows and analyzes ten optimal CCTS supply chain designs, each complying with a different target of CO_2 emissions ranging from the minimum-emissions design (left-hand side of Fig. 2, corresponding to low values of the total system emissions on the x-axis) to a design capturing about 10% of the total WtE emissions only (right-hand side of Fig. 2, high values of emissions). Each CCTS supply chain design corresponds to a stand-alone configuration, optimized independently of the others. With respect to total system costs and emissions, these designs lay on a Pareto front bracketed between two extreme points: (i) the BAU configuration, where no CO_2 is captured from the WtE plants, total system costs are zero, and total system emissions are maximum and equal to ca. 4.5 $MtCO_2/y$ (not shown in Fig. 2 for the sake of simplicity), and (ii) the minimum-emissions configuration (0.54 $MtCO_2/y$ of total system emissions), whereby CO_2 is captured at all thirty WtE plants with a maximum CO_2 capture rate of 90%, resulting in maximum total system costs.

Three panels are displayed in Fig. 2: panel (a) plots the amount of CO_2 stored, split in its fossil and biogenic fractions (for each CCTS supply chain design), and the amount of net-negative emissions generated (i.e., the difference between the biogenic emissions stored and the emissions caused by energy consumption upon CO_2 capture, transport and storage); panel (b) shows the economic indicators, namely total system costs and costs of stored and avoided carbon, split in the three contributions due to transport (the largest), capture, and storage (the smallest); panel (c) plots the lengths of the transport connections installed for the different transport modes and the number of capture units installed. Additionally, Fig. 3 shows a schematic representation of the CCTS supply chain corresponding to four prototypical designs, denoted as D1, D3, D5, and D8 in Fig. 2. Both Figs. 3 and 2(c) highlight the structural changes of the CO_2 network as the amount of CO_2 managed (i.e., captured, transported and stored) increases from design D1, to D3, to D5, and to D8.

The CO_2 emissions stored and the total system costs increase monotonically for decreasing total system emissions, i.e., for increasing amounts of CO_2 captured, transported and stored. For example, a reduction in system emissions of ca. 8% with respect to BAU results in total costs of ca. 136 MEUR (D1), while a reduction of ca. 88% leads to total costs of ca. 590 MEUR (D10), i.e., three times larger. This reduction corresponds to the minimum attainable value of CO_2 emissions, obtained when CO_2 capture units with a recovery rate of 90% are installed in all WtE plants, and considering the emissions generated by the CCTS supply chain. Although additional emissions are produced to operate the CCTS supply chains, these are within 10% of the total emissions stored and allow to generate up to 2.0 $MtCO_2/y$ of net-negative emissions.

The cost indicators CSC and CAC generally decrease for increasing amounts of CO_2 emissions managed by the network as a result of economies of scale achieved through the sharing of infrastructure by multiple industrial emitters. However, exceptions to this trend consist of designs relying on a transport infrastructure that is not effectively deployed, see D2. In this case, a relatively large investment is needed to transport only a limited amount of CO_2, required to meet the emissions target, resulting in an increase of CSC and CAC with respect to D1. The minimum CSC and CAC values attained correspond to ca. 150 EUR/tCO_2 (see D8 and D10). Furthermore, the ratio between CSC and CAC is larger than 0.9 and also decreases for increasing amounts of CO_2 managed; this suggests that the additional emissions caused by CO_2 capture and

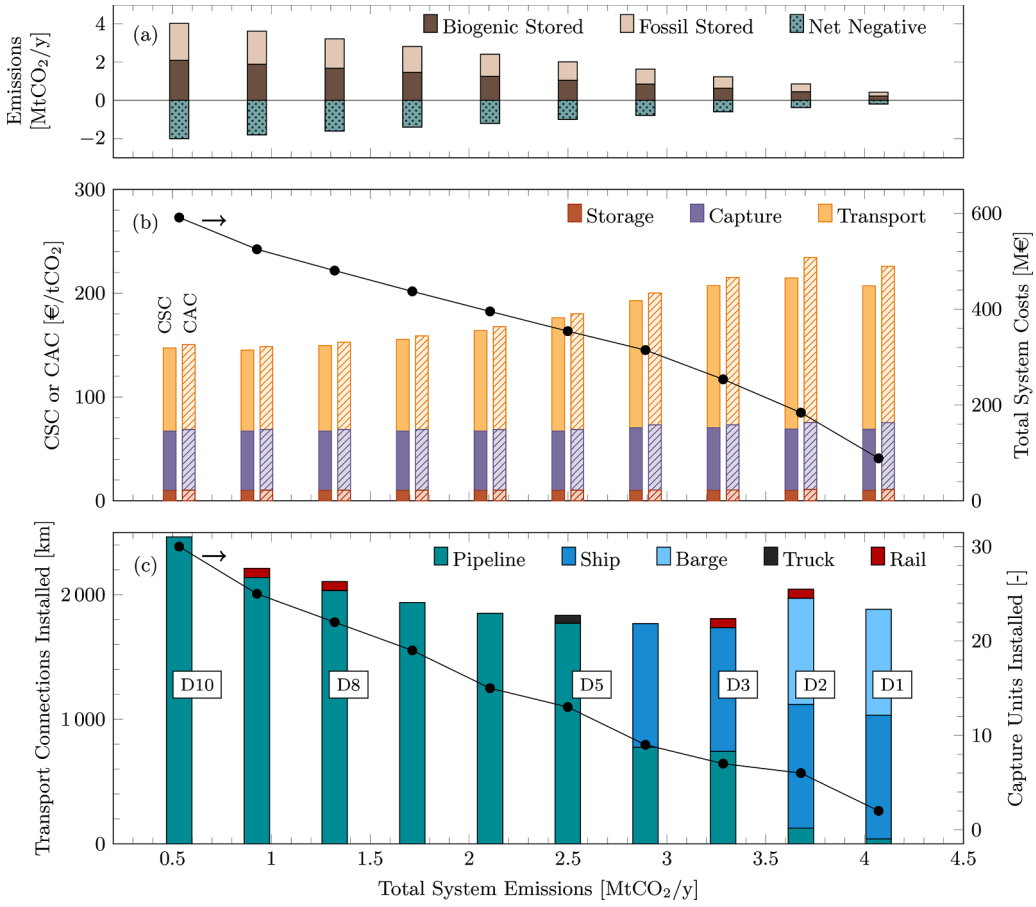


Fig. 2. Static designs of the CCTS supply chain lying on a cost-emissions Pareto front: (a) biogenic and fossil CO_2 emissions stored, and net-negative emissions generated, (b) breakdowns of CSC and CAC, and total system costs, (c) total length of connections installed for each CO_2 transport type and number of installed CO_2 capture units as function of the total system emissions.

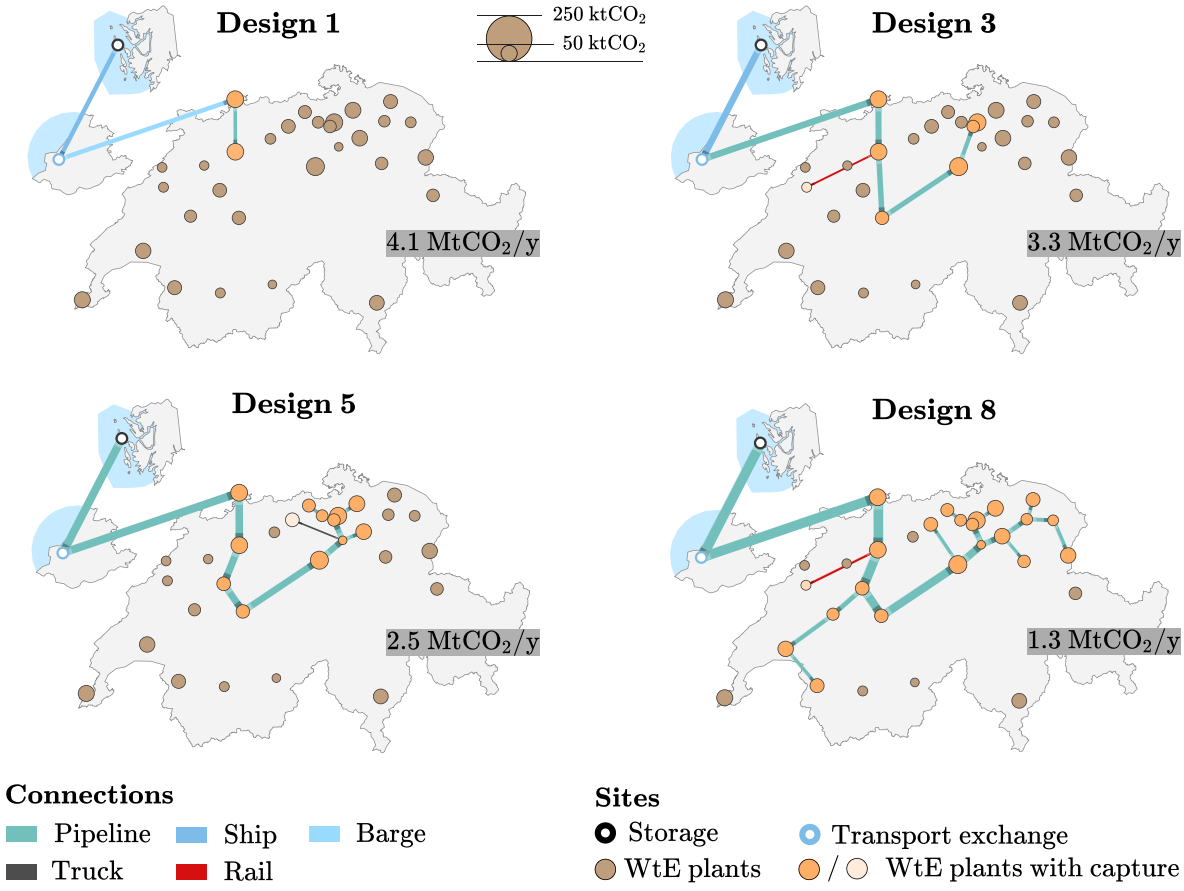


Fig. 3. Structures of four cost-optimal CCTS supply chain designs satisfying different CO_2 emissions targets, corresponding to a CO_2 emissions reduction of about 8% (D1), 26% (D3), 56% (D5), and 71% (D8) with respect to BAU. *Note:* The light-shaded yellow dots indicate (e.g., in D3, D5, and D8) indicate that the corresponding WtE plants are capturing CO_2 with a recovery rate smaller than 90%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

transport per unit CO_2 managed are generally limited, and decrease in relative terms as larger volumes of CO_2 are handled, thus leading to a more efficient network. The largest contribution to costs is due to CO_2 transport (ca. 60% on average), followed by CO_2 capture (ca. 34%), and CO_2 storage (ca. 6%).

Reductions in total system emissions are achieved primarily by installing additional CO_2 capture units, and secondarily by progressively selecting CO_2 transport modes with reduced carbon footprint as the amount of CO_2 managed increases. For example (see Fig. 3), the barge connection between Basel and Rotterdam (see D1) and the ship connection between Rotterdam and the Northern Lights storage site (see D1 and D3) are replaced by pipeline connections (characterized by a smaller environmental impact) as the emissions target becomes more stringent, thus resulting in higher costs (see D5 and D8). Rail and truck connections are selected only for small CO_2 flow rates and short distances (less than ca. 75 km) as they provide low cost connections but deliver only limited emissions reductions (see D3, D5, and D8 in Fig. 3). In other words, for specific designs, the system can *afford* to utilize connections with a bigger environmental impact (i.e., rail and truck) to satisfy the emissions reduction target imposed, while profiting from their lower costs. Similarly, some CCTS supply chain designs consists of CO_2 capture units operating at a recovery rate lower than 90%, which enables satisfying the required emissions target while limiting total costs (see D3, D5, and D8 in Fig. 3). Finally, the CCTS supply chain design enabling minimum emissions consists exclusively of pipeline connections (see Fig. 2(c), D10).

5. Dynamic design of CCTS supply chains

5.1. Linear emissions reduction (LER) pathway

Figs. 4–6 illustrate the cost-optimal time evolution of the CCTS supply chain design complying with a LER pathway over the considered time horizon. Fig. 4 shows the following variables as function of the year along the time horizon: (a) the amount of CO_2 stored, split in its fossil and biogenic fractions, and of net-negative emissions generated; (b) the total system costs, split in the three contributions due to transport, capture, and storage; (c) the economic indicators, LCSC and LCAC, split in the three contributions due to transport, capture, and storage. Fig. 5 plots: (a) the number of capture units installed; and (b) the total lengths of connections installed and the amount of CO_2 transported per each CO_2 transport mode as function of the year along the time horizon. Additionally, Fig. 6 shows a schematic representation of the CCTS supply chain design corresponding to four given years along the time horizon.

The minimum emissions level of ca. 0.54 $MtCO_2/y$ is attained in the last year, i.e., year 25. Accordingly, the amount of CO_2 emissions managed increases linearly up to ca. 4.0 $MtCO_2/y$, thus resulting in about 1.5 $MtCO_2$ of net-negative emissions in the last year. This scenario assumes that a Swiss storage site becomes available with unlimited capacity after 15 years into the 25 years time horizon.

The total system costs increase monotonically over the first 15 years as the CCTS supply chain is expanded to capture, transport and store larger amounts of CO_2. The total costs drop by nearly 40% when the

Swiss CO_2 storage site becomes operational, owing to the decrease in transport costs as the CO_2 is rerouted to the nearer Swiss storage site.

Over the full time horizon (i.e., in year 25), the LCSC and LCAC attain values of 168 and 174 EUR/tCO_2, respectively. The levelized costs initially increase as large investments are made to store/avoid only limited amounts of CO_2: over the first five years total system costs of ca. 482 MEUR are required to store a total amount of 2.6 MtCO_2. Nevertheless, a decreasing trend is obtained in the remaining years mostly due to the greater amount of CO_2 handled, hence to the better exploitation of the transport network as a shared infrastructure, and partly due to the utilization of the closer Swiss CO_2 storage site.

The number of CO_2 capture units installed increases over the time horizon, and in year 25 all WtE plants capture CO_2 at the maximum recovery rate of 90%. Until year 4, the international transport of the captured CO_2 relies on barge and ship connections between Basel-Rotterdam and Rotterdam-Northern Lights, respectively. Although the barge connection covers a long distance (ca. 850 km), it is deployed only for limited amounts of CO_2 transported (<0.68 MtCO_2/y), and it is replaced by a pipeline connection starting from year 5. After year 15, when the Swiss storage site becomes operational, the CO_2 is rerouted to the domestic storage site. The connections to the North Sea storage site through Rotterdam are no longer utilized for transporting CO_2 captured at the Swiss WtE plants, but remain available, possibly for serving the needs of other CO_2 emitters, until the end of the considered time horizon (i.e., the investment contribution is paid over the entire time horizon). Furthermore, during the first 15 years the CO_2 network develops mainly within Northeastern Switzerland, which has a higher density of WtE plants clustered in a relatively small area. Only in the last ten years, the CO_2 network branches out to the Western and the Southern regions of Switzerland, which have fewer WtE plants, located at larger distances between each other.

By year 25, the CO_2 transport network is based exclusively on

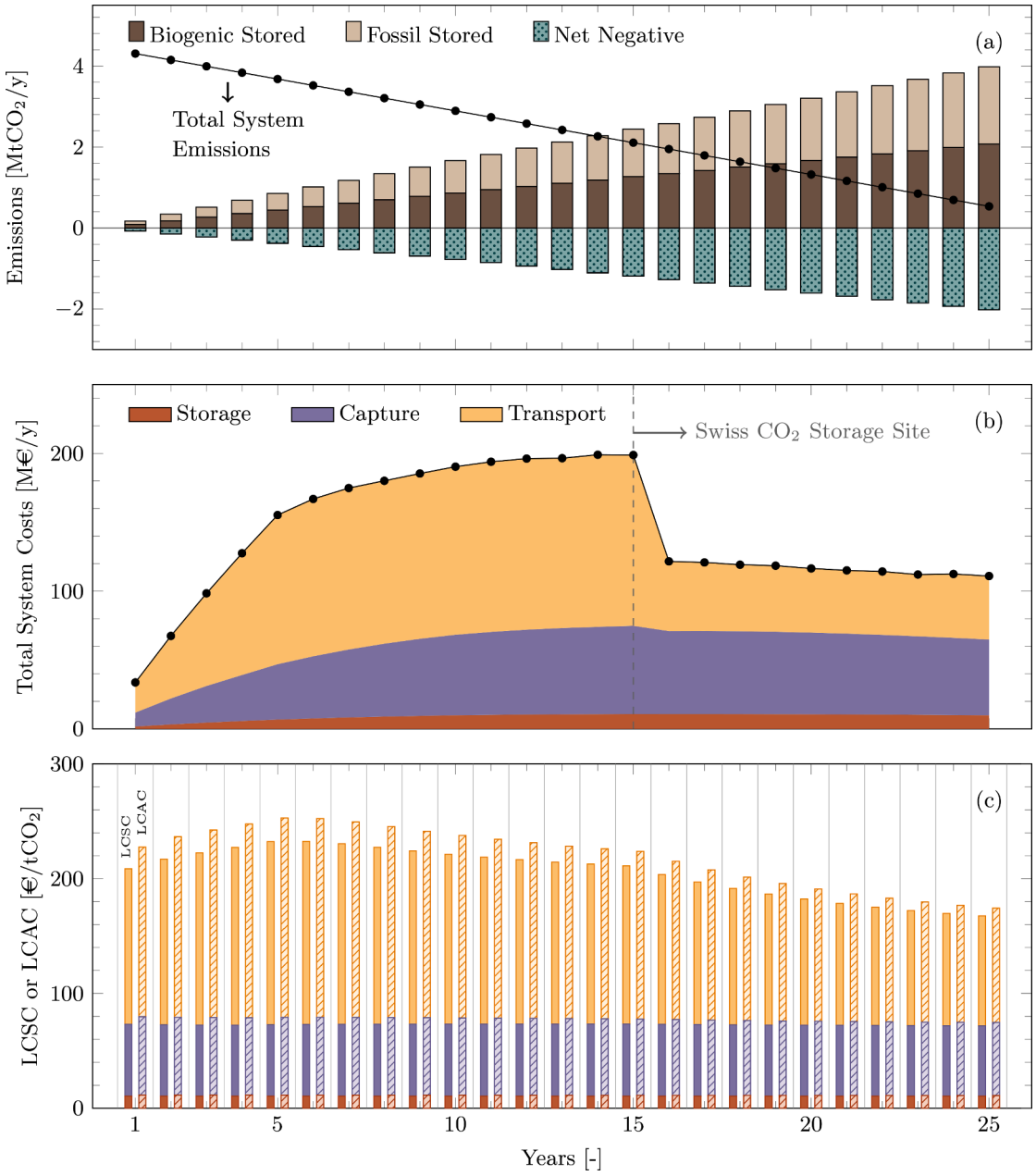


Fig. 4. Cost-optimal evolution of the CCTS supply chain complying with a LER pathway over a 25-year time horizon: (a) total system emissions, emissions stored, and net-negative emissions generated, (b) breakdown of total system costs, (c) breakdown of LCSC (solid bars) or LCAC (dashed bars) as function of the year of the time horizon. The Swiss CO_2 storage site becomes operational after year 15.

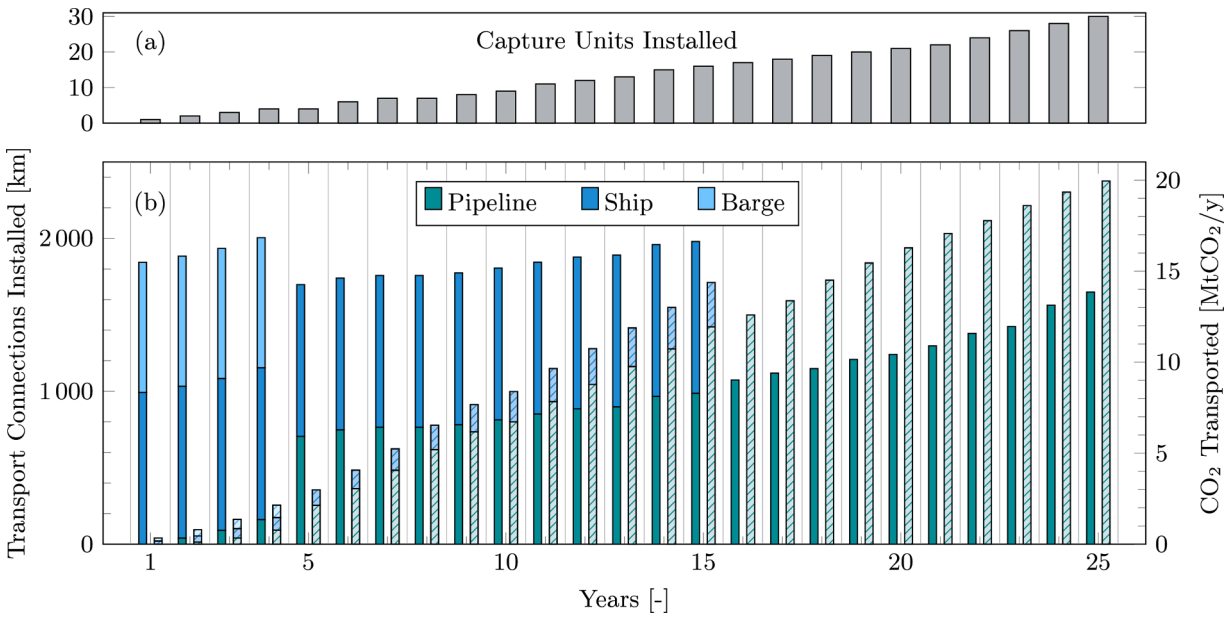


Fig. 5. Cost-optimal evolution of the CCTS supply chain complying with a LER pathway over a 25-year time horizon: (a) number of capture units installed, and (b) total length of connections installed (solid bars) and amount of CO_2 transported (dashed bars) per each CO_2 transport type, i.e., total deployed capacity, as function of the year of the time horizon.

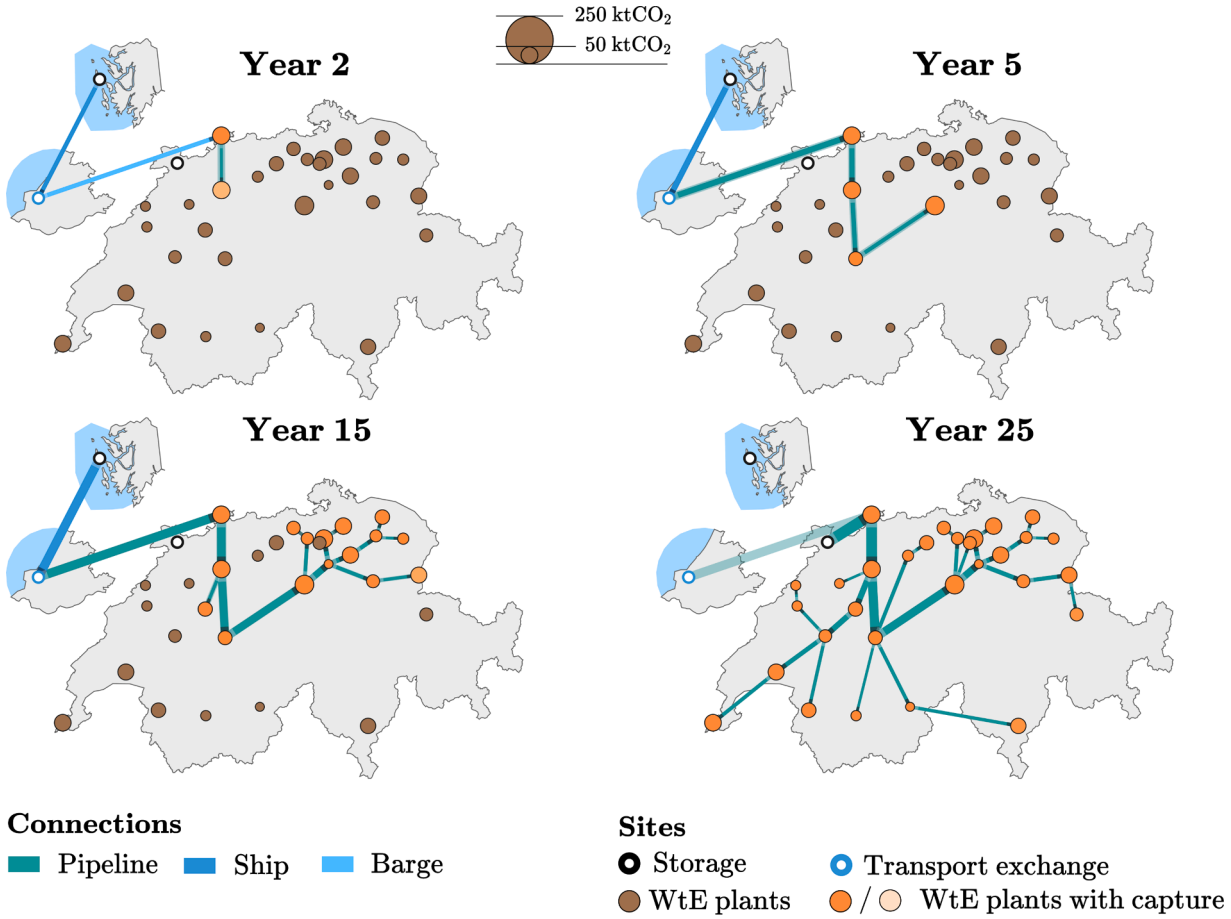


Fig. 6. Cost-optimal evolution of the CCTS supply chain complying with a LER pathways over a 25-year time horizon: Schematic representation of the optimal structure of the supply chain in different years, namely year 2, 5, 15, 25. *Note*: The pipeline connection between Basel and Rotterdam in year 25 represented with a shaded colour indicates that the connection is in place but it is no longer utilized to transport CO_2 captured at the Swiss WtE plants (i.e., the CO_2 flow through the connection is zero, while its size is not, see Fig. 1); the light-shaded yellow dot in year 1 indicates that the corresponding WtE plant is capturing CO_2 with a recovery rate smaller than 90%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

pipelines, which stand out as the main component of cost-effective CCTS supply chains. Despite their high investment costs, which scale linearly with distance and are much more affected by distance than capacity, the cost per unit CO_2 transported drops rapidly when operating at higher flows. In fact, in year 25, the transport costs amount to roughly 0.01 EUR/(tCO_2 km). Nevertheless, compared to pipelines, other transport modes, such as ship and barge, offer higher flexibility and lower investment costs, thus making them particularly suitable for temporary connections that are needed while the CCS infrastructure is being developed and new storage sites become operational. This is the case of the ship connection between Rotterdam and the Northern Lights storage site, which transports ca. 1.0 $MtCO_2$ in year 6. For this volume of CO_2, the yearly operation costs for a ship far exceed those of the pipeline. However, the optimizer leverages the lower investment cost alternative, which is provided by the ship, to fulfill the demand over a shorter time period and to produce a more cost-effective solution.

Finally, in a mature CCTS industry relying on CO_2 storage in the North Sea, the international CO_2 pipelines infrastructure would be shared by multiple European emitters, thus most likely leading to a reduction of the LCSC.

5.2. Cumulative emissions reduction (CER) pathway

Figs. 7–9 illustrate the cost-optimal time evolution of the CCTS supply chain complying with the cumulative emissions reduction pathway over the considered time horizon, with plots analogous to those presented in the previous section. Fig. 7 shows the following variables as function of the year along the time horizon: (a) the amount of CO_2 stored, split in its fossil and biogenic fractions, and of net-negative emissions generated; (b) the total system costs, split in the three contributions due to transport, capture, and storage; (c) the economic indicators, LCSC and LCAC, split in the three contributions due to

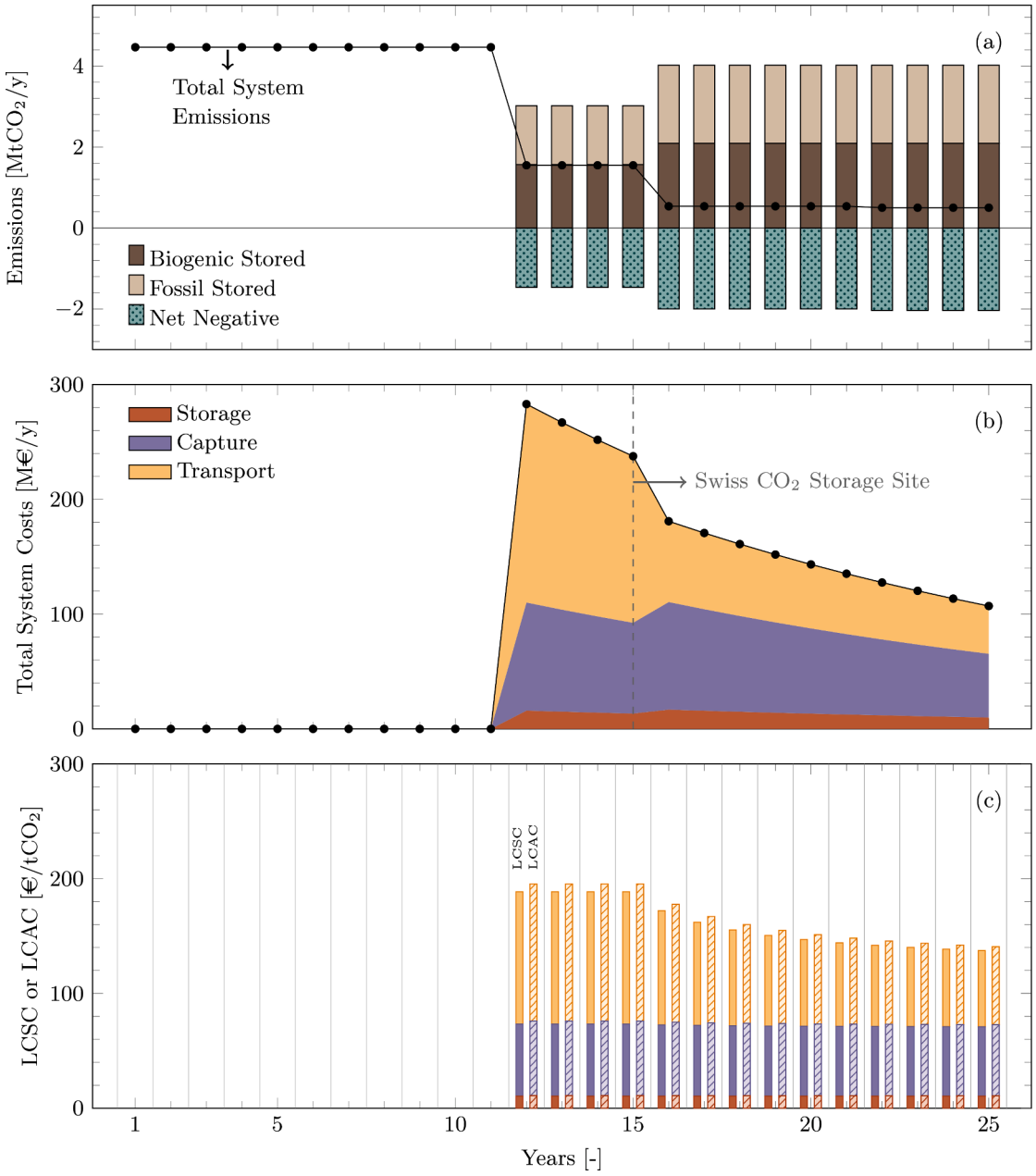


Fig. 7. Cost-optimal evolution of the CCTS supply chain complying with a CER pathway over a 25-year time horizon: (a) total system emissions, emissions stored, and net-negative emissions generated, (b) breakdown of total system costs, (c) breakdown of LCSC (solid bars) or LCAC (dashed bars) as function of the year of the time horizon. The Swiss CO_2 storage site becomes operational after year 15.

transport, capture, and storage. Fig. 8 displays: (a) the number of capture units installed, and (b) the total lengths of connections installed and amount of CO_2 transported per each CO_2 transport mode as function of the year along the time horizon. Additionally, Fig. 9 shows a schematic representation of the CCTS supply chain corresponding to two given years along the time horizon.

No CCTS supply chain is installed during the first 11 years of the time horizon considered, hence resulting in BAU emissions, which are then reduced by 65% in year 12 and by 88% in year 16, as the domestic storage site becomes operational. Similarly, the amount of CO_2 managed follows a step-like trend, with ca. 3.0 and 4.0 $MtCO_2/y$ being stored during years 12 to 15 and 16 to 25, respectively.

From an economic perspective, delaying the installation of CO_2 capture units and transport network towards the end of the time horizon appears advantageous due to two factors: (i) the discounting of future cash flows, and (ii) the possibility after 15 years of exploiting the closer domestic storage site, thus leading to lower transport costs. The influence of these two factors is visible in the total system costs, which follow approximately linear trends between years 12 and 15 and between years 16 to 25, and exhibit a clear drop of 24% in year 16, as observed for the LER pathway. When following the CER pathway, the cost reduction due to the domestic storage site becoming operational (i.e., between year 15 and year 16) is limited by the steep increase in amounts of CO_2 captured and in the corresponding capture costs. Compared to LER, the CER pathway results in lower LCSC and LCAC, which are equal to 137 and 141 EUR/tCO_2, respectively.

Consistently, during the first four years of operation of the CCTS supply chain (from year 12 to year 15), 21 WtE plants are capturing and transporting CO_2 for geological storage, and by year 16 all 30 plants are capturing CO_2 at the maximum recovery rate. Therefore, there is no progressive expansion of the CCTS supply chain over time, but rather this is installed in two steps. In year 12, all the captured CO_2 is

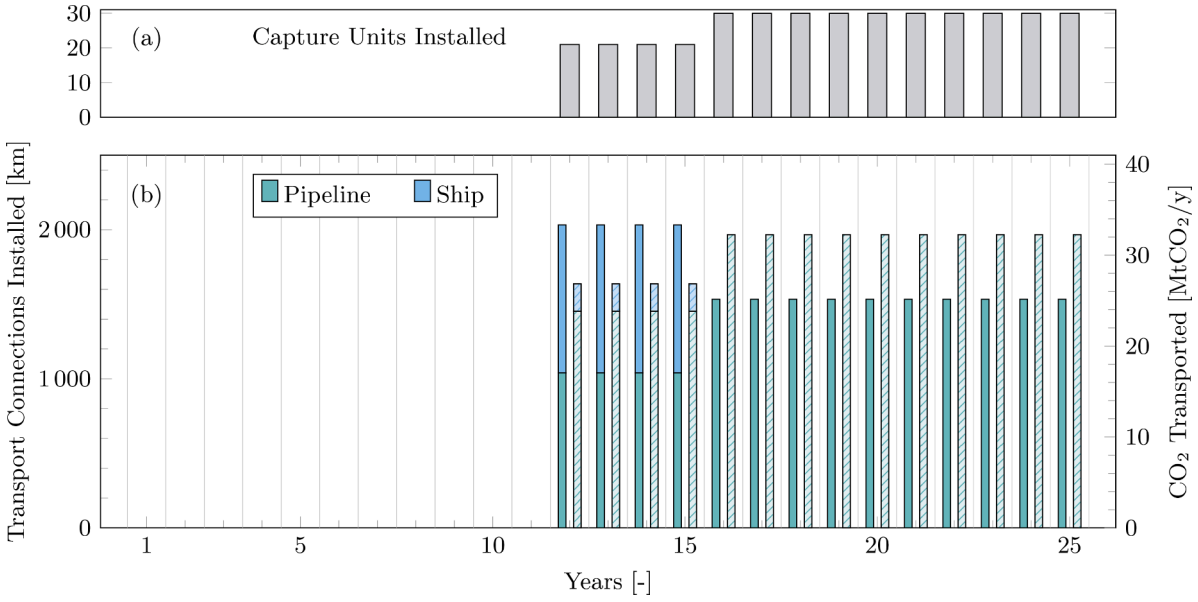


Fig. 8. Cost-optimal evolution of the CCTS supply chain complying with a CER pathways over a 25-year time horizon (from the top): (a) number of capture units installed, and (b) total length of connections installed (solid bars) and amount of CO_2 transported (dashed bars) per each CO_2 transport type, i.e., total deployed capacity, as function of the year of the time horizon.

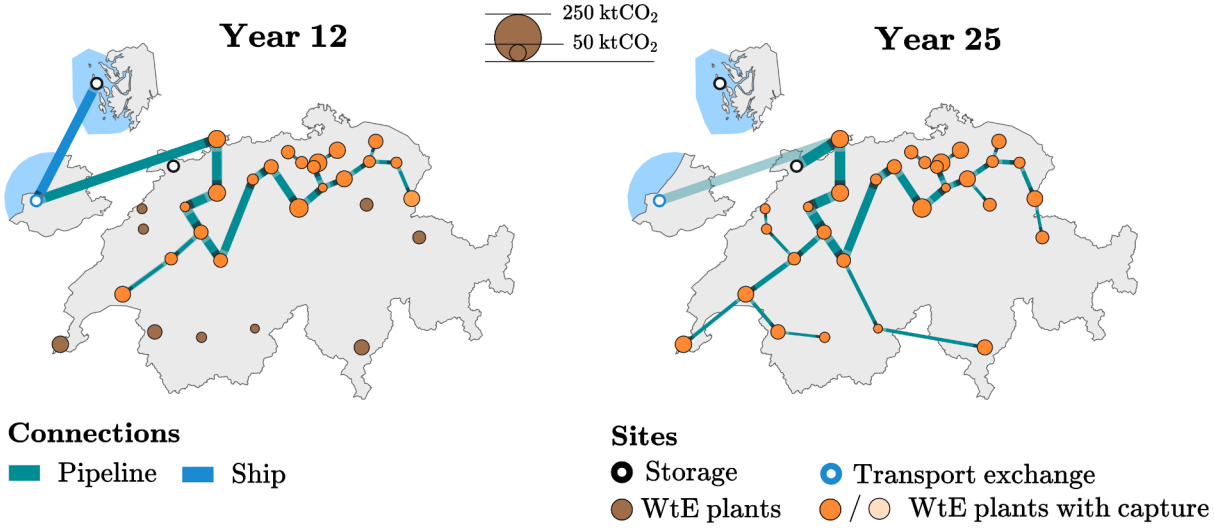


Fig. 9. Cost-optimal evolution of the CCTS supply chain complying with a CER pathway over a 25-year time horizon: Schematic representation of the optimal structure of the supply chain in different years, namely year 12 and 25. *Note*: The pipeline connection between Basel and Rotterdam in year 25 represented with a shaded colour indicates that the connection is in place but it is no longer utilized to transport CO_2 captured at the Swiss WtE plants.

transported by pipelines (for a total length of ca. 1040 km) to Rotterdam, and from there by ship to the Northern Lights storage site. In year 16, the CO_2 network is extended to collect CO_2 from all WtE plants and deliver it to the domestic storage site. Afterwards the network design remains unchanged and relies entirely on pipelines, as they constitute the most cost-effective and environment-friendly transport solution.

Nevertheless, the optimal solution following a CER pathway is the outcome of a purely techno-economic optimization. In practice, real-life circumstances and risks associated with a waiting strategy (e.g., setbacks in the construction of a domestic storage site, sub-optimal rollout of the CO_2 capture units due to different decisions of the WtE plant operators) may negatively affect the presumed economic and environmental advantages stemming from delaying actions in implementing CCTS chains.

5.3. Impact of start-of-operation year of domestic storage site

The results presented in Sections 5.1 and 5.2 illustrate the economic advantage of exploiting a CO_2 storage site located at a comparatively short distance from the industrial emitters. The influence of the year in which the domestic storage site becomes operational is investigated through a sensitivity analysis for the LER and CER pathways, see Fig. 10. This analysis provides insights both on the economic advantage for CO_2 emitters of having readily available a CO_2 storage site located nearby (e.g., in Switzerland or at a short distance in a neighboring country) and on the shortcomings of having available such a storage only at a later time in the future.

Let us first analyse the LER scenario. Were a domestic storage site (located at an average distance of 165 km from all considered emitters) operational from the beginning of the time horizon, then the LCSC would drop below 100 EUR/tCO_2; this would correspond to a decrease of ca. 50% with respect to the LCSC achieved if only the Northern Lights storage site (located at an average distance of 1500 km from all considered emitters) were available. Delaying the start of operation of the domestic storage site from year 1 to year 25 increases the LCSC from less than 100 EUR/tCO_2 to more than 180 EUR/tCO_2, according to the trend shown in Fig. 10a. LCSC and LCAC follow a similar trend, with the difference between the two increasing for later start-of-operation of the domestic CO_2 storage site. This is because of the greater CCTS supply chains emissions, caused by the greater distance between the CO_2 sources and the storage site.

Similar considerations apply to the CER pathway. However, in this case, the start of operation of the domestic storage site has no impact if it occurs within the first 10 years, since the CCTS supply chains start being installed from year 11 in any case (see Figs. 7 and 8). Overall, similar values of LCSC and LCAC are obtained for LER and CER pathways, though smaller values are consistently obtained for the latter thanks to the higher planning flexibility. Consistently, the difference in costs between the two pathways increases for a later start-of-operation, because of the greater impact of planning flexibility for CCTS supply chains that are spread over longer CO_2 transportation distances.

6. Conclusions

This work presents a novel optimization framework for the optimal design of CCTS supply chains in terms of installation, sizing and operation of CO_2 capture and transport technologies. The optimal design problem is formulated as a MILP that minimizes the total costs of the supply chains while complying with different emissions reduction pathways over a time horizon of 25 years. All design decisions are time-dependent and are taken with a yearly resolution. Alongside the dynamic multi-year evolution of CCTS supply chains, the static, cost-optimal design of CCTS supply chains is also investigated for different targets of total system CO_2 emissions.

The model is flexible, general, and suited to be applied to a variety of hard-to-abate industrial sectors, as well as different geographic locations and time frames. Here it is illustrated by investigating the optimal design of a CCTS supply chain to decarbonize the Swiss WtE sector over 25 years, e.g., from 2025 to 2050, where two potential CO_2 storage sites are considered, one in the North Sea, which will be operational starting from mid-2020s, and a hypothetical one in Switzerland, potentially available in the future. The CO_2 is captured at the WtE plants and is transported to the storage sites via several possible modes of transport, which include pipeline, truck, rail, barge, and ship.

Cost-optimal CCTS supply chains are designed according to two emissions reduction pathways, namely a linear (LER) and a cumulative emissions reduction (CER) pathway. The former implies a limited planning flexibility, where the CCTS supply chains must comply with a specified target evolution of yearly emissions. Such evolution in yearly emissions goes from the current emissions without CCTS at the beginning of the time horizon to the minimum attainable emissions at the end of the time horizon. The latter implies a greater planning flexibility, where the CCTS supply chains must only comply with the cumulative emissions target over the entire time horizon and with the minimum attainable emissions at the end of the time horizon. For equal emissions reduction, the greater flexibility of the CER pathway enables lower costs of stored and avoided carbon, i.e., about 15% smaller, than those

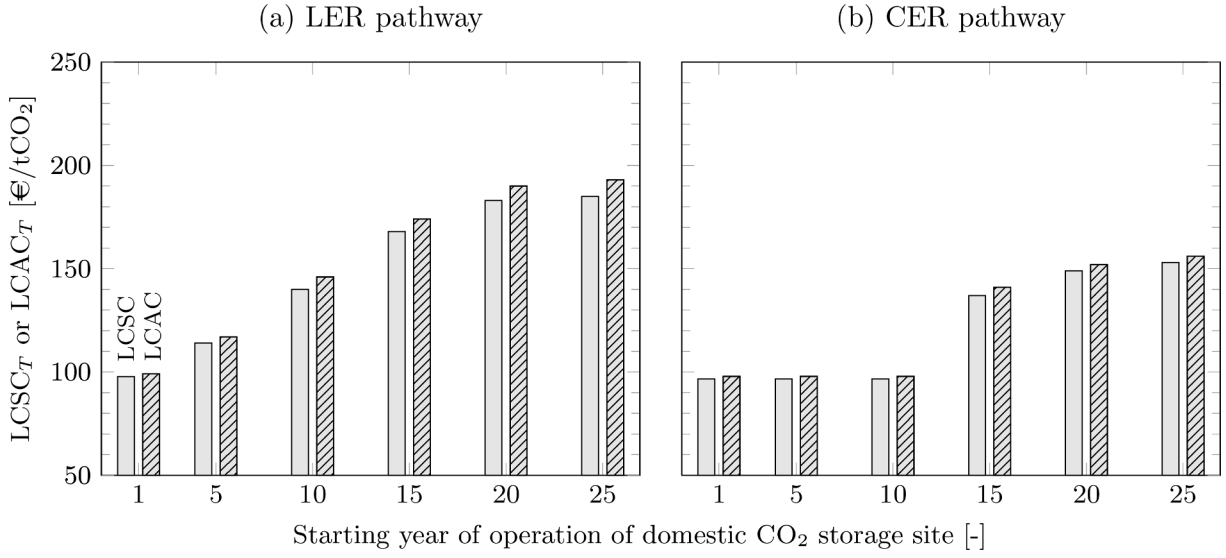


Fig. 10. $LCSC_T$ and $LCAC_T$ as functions of the year after which the Swiss CO_2 storage site becomes operational, for (a) LER and (b) CER pathways.

accrued along the LER pathway.

Findings show that pipelines are the most cost-effective mode of transport when large volumes of CO_2 (around 1.5 $MtCO_2$/y in this case) must be handled, especially when considering multi-year time horizons for the planning of CCTS supply chains. Ship and barge connections are competitive with pipeline connections, whereas rail and truck connections are cost effective only when considering limited planning horizons or small volumes of CO_2 to be transported. Furthermore, results show that for a country like Switzerland, and generally for any region that is relatively far (more than ca. 1000 km away) from soon-available storage sites, the cost of CO_2 transport represents the greatest contribution to the overall costs of CCTS supply chains. Such a transport cost is greatly reduced and becomes similar to or lower than the capture cost for CO_2 storage sites located nearby, i.e., less than ca. 200 km away, and particularly for domestic sites.

From a broader perspective, it is worth underlining that the analysis of the two scenarios, i.e., LER and CER, has highlighted important differences, in terms of timing, structure, and cost of the CO_2 network, associated to the two different climate policies. The CER policy yields more flexibility and cost reductions with respect to the LER policy. But, on the flip side, it is less robust and may lead, in case of delays, to failure in achieving the climate goals, because of lack of time for corrective actions.

In both cases though, we have assumed that once the climate policy is decided, the CO_2 network rollout develops in an optimal way from an "unbiased" economic and financial perspective. Reality may well be different, with the timing and features of the rollout being dominated by local political and financial decisions, which may be completely unrelated to considerations of overall costs reduction. Though the study of such scenario(s) is beyond the scope of this work, it is worth noting that the network model and the optimization approach presented here are certainly suited for the quantitative study of different scenarios, where additional constraints other than those dictated by a centralized climate policy are considered and enforced.

CRediT authorship contribution statement

Viola Becattini: Conceptualization, Methodology, Data curation, Writing – original draft, Visualization, Project administration. **Paolo Gabrielli:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Project administration. **Cristina Antonini:** Methodology, Software, Writing – original draft. **Jordi Campos:** Methodology, Software, Visualization. **Alberto Acquilino:** Software. **Giovanni Sansavini:** Writing – review & editing, Supervision. **Marco Mazzotti:** Conceptualization, Methodology, Writing – review & editing, Visualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Detailed formulation of the optimization problem

All aspects of the optimization problem introduced in Section 3 of the paper, namely input data, decision variables, constraints, and objective function, are described in detail in the following. The following description builds upon the general model formulation introduced earlier (Gabrielli et al., 2020; 2019; 2018a; 2018b) and describes the specific case of CCTS supply chains.

Concerning notation, bold symbols indicate vectors or matrices. The set of available transport technologies is indicated with $\mathscr{L}$, and the set of nodes with $\mathscr{N}$; T is the length of the time horizon. Here, $\mathscr{L}$ includes pipeline, truck, rail, ship and barge; $\mathscr{N}$ includes the location of all WtE plants in Switzerland, of the Swiss storage site, of the continental terminal in the Rotterdam harbor, and the Northern Lights storage site. The subset of nodes that includes the location of the WtE plants, i.e. the capture sites, is denoted as $\mathscr{N}_C$, and the one that includes the location of the storage sites is denoted as $\mathscr{N}_S$. Unless otherwise indicated, the quantities below are non-negative real numbers.

A1. Input data

Inputs to the optimization problem are:

• CO_2 emissions of WtE plants	E_i	$\forall i \in \mathscr{N}$
• Capacity of CO_2 storage sites	$\boldsymbol{A}_i \in \mathbb{R}^T$	$\forall i \in \mathscr{N}$
• Electricity price	u	
• Electricity carbon intensity	ϵ_i	$\forall i \in \mathscr{N}$
• Availability of CO_2 transport technologies	$\boldsymbol{w}_{i,i',l} \in \{0,1\}^T$	$\forall i, i' \in \mathscr{N}, \forall l \in \mathscr{L}$
• Node distance for CO_2 transport technologies	$L_{i,i',l}$	$\forall i, i' \in \mathscr{N}, \forall l \in \mathscr{L}$
• System configuration (e.g. emissions reduction scenario - see Section 2 of the paper)		
• Cost and performance for CO_2 capture, conditioning and transport		

Here, the subscript "i,i'" indicates a connection going from node i to node i'; $E_i = 0$ in transit and storage nodes, whereas $A_{i,t} = 0$ in transit and capture nodes for all $t \in \{0, \ldots, T\}$.

The CO_2 emissions of WtE plants is taken from the Swiss Emissions Trading Registry (Swiss Federal Office of the Environment, 2021) and is reported in Table B.4, while the information on the biogenic share of emissions (ca. 52%) is taken from the report by the Association of Operators of Swiss Waste Recycling Plants (VBSA, 2019). The distances for road, rail and barge transport are calculated using the API of Google maps (Google, 2021), whereas distances for ship and pipeline transport are based on direct paths (i.e., as-the-crow-flies distances). All numerical values used to perform the optimizations are reported and discussed in Appendix B.

A2. Decision variables

The values of the following decision variables are determined by the optimization procedure:

• Installation of CO_2 capture units	$\boldsymbol{x}_i \in \{0,1\}^T$	$\forall i \in \mathscr{N}$
• Size of CO_2 capture units	$\boldsymbol{S}_i \in \mathbb{R}^T$	$\forall i \in \mathscr{N}$
• Installation of CO_2 transport technologies	$\boldsymbol{y}_{i,i',l} \in \{0,1\}^T$	$\forall i, i' \in \mathscr{N}, \forall l \in \mathscr{L}$
• Size of CO_2 transport technologies	$\boldsymbol{R}_{i,i',l} \in \mathbb{R}^T$	$\forall i, i' \in \mathscr{N}, \forall l \in \mathscr{L}$
• Flow of CO_2 transport technologies	$\boldsymbol{G}_{i,i',l} \in \mathbb{R}^T$	$\forall i, i' \in \mathscr{N}, \forall l \in \mathscr{L}$
• Scheduling of CO_2 transport technologies	$\boldsymbol{z}_{i,i',l} \in \{0,1\}^T$	$\forall i, i' \in \mathscr{N}, \forall l \in \mathscr{L}$
• CO_2 captured at WtE plants	$\boldsymbol{P}_i \in \mathbb{R}^T$	$\forall i \in \mathscr{N}$
• CO_2 stored at storage sites	$\boldsymbol{U}_i \in \mathbb{R}^T$	$\forall i \in \mathscr{N}$
• Electricity consumed for capture / storage	$\boldsymbol{F}_i \in \mathbb{R}^T$	$\forall i \in \mathscr{N}$
• Electricity consumed for CO_2 conditioning	$\boldsymbol{V}_i \in \mathbb{R}^T$	$\forall i \in \mathscr{N}$

The electricity consumed is imported from the electricity grid, which is assumed to be available at all nodes.

A3. Constraints

The constraints of the optimization problem can be grouped into two categories, namely the CO_2 mass balances and the constitutive equations for the individual capture, transport and storage technologies.

- CO_2 mass balances. For all nodes $i \in \mathscr{N}$ and for all years $t \in \{0, \dots T\}$, the CO_2 mass balance is expressed as:

$$P_{i,t} + \sum_{l \in \mathscr{L}} \sum_{i' \in \mathscr{N}, i \neq i'} \left[G_{i',i,l,t} \left(1 - \lambda_l L_{i,i',l}\right) - G_{i,i',l,t} \right] - U_{i,t} = 0 \tag{A.1}$$

where λ_l is a parameter quantifying the loss of CO_2 when transported through transport technology l and $L_{i,i',l}$ is the distance between nodes i and i' when using transport technology l. The expression of CO_2 loss follows the differential equation

$$\frac{\mathrm{d}G}{\mathrm{d}L} = -\lambda G \tag{A.2}$$

which returns

$$G = G_0 \exp(-\lambda L) \approx G_0(1 - \lambda L), \quad \text{for } \lambda L \to 0 \tag{A.3}$$

- Constitutive equations for capture, transport and storage technologies.

Capture technologies. For all capture units, the size must vary between zero (i.e. the capture unit is not installed) and the maximum value attained when considering the maximum capture rate, i.e. $\psi = 0.9$, which corresponds to capturing 90% of the CO_2 emissions of the corresponding WtE plant. For all nodes $i \in \mathscr{N}$ and for all years $t \in \{0, \dots T\}$, this is expressed as:

$$0 \leq S_{i,t} \leq \psi E_i x_{i,t} \tag{A.4}$$

Within the proposed MILP framework, the performance of the CO_2 capture plant is described through a linear expression correlating the amount of CO_2 captured (i.e. the output stream of the capture unit, $P_{i,t}$) with the required electricity (i.e. the input stream of the capture unit, $F_{i,t}$). For all nodes $i \in \mathscr{N}$ and for all years $t \in \{0, \dots T\}$:

$$P_{i,t} = \eta_i F_{i,t} \tag{A.5}$$

where,

$$0 \leq P_{i,t} \leq S_{i,t} \tag{A.6}$$

Here, η_i is the conversion efficiency quantifying the amount of CO_2 captured per unit electricity consumed by the WtE plant in node i (Roussanaly et al., 2020).

Storage technologies. The CO_2 storage sites are modeled as technologies that use electricity to store CO_2, characterized by a conversion efficiency, η, which accounts for the compression expenditure required for injection:

$$U_{i,t} = \eta_i F_{i,t} \tag{A.7}$$

where,

$$0 \leq U_{i,t} \leq A_{i,t} \tag{A.8}$$

Transport technologies. The possibility of installing a transport technology $l \in \mathscr{L}$ is constrained by the availability of the corresponding connection between nodes i and i', with $i, i' \in \mathscr{N}$ and $i \neq i'$, while the installed size is constrained by the minimum and maximum available values for that connection:

$$y_{i,i',l,t} R^{\min}_{i,i',l} \leq R_{i,i',l,t} \leq y_{i,i',l,t} R^{\max}_{i,i',l} \tag{A.9}$$

where,

$$y_{i,i',l} \leq w_{i,i',l} \tag{A.10}$$

Here, the size of a transport technology is given by its rated mass flow rate.

The flow of CO_2 for all connections is constrained by the installed size and by the minimum attainable value:

$$z_{i,i',l,t} G^{\min}_{i,i',l} \leq G_{i,i',l,t} \leq z_{i,i',l,t} R_{i,i',l,t} \tag{A.11}$$

Note that the bilinearity between the continuous variable R and the binary variable z can be recast as two linear constraints (Glover, 1975).

Energy required for conditioning. CO_2 is produced and transported in different states and at different levels of pressure and temperature. Here, this is taken into account by considering the electrical energy, V, required for compressing, cooling and liquefying the CO_2 when produced and transported via different technologies (see Table B.1 in Appendix B). This is modeled by considering that in each node the CO_2 is first brought to some reference conditions, which correspond to the capture conditions, and then brought to the transport conditions. For all nodes $i \in \mathscr{N}$, the electricity consumption required for conditioning is then expressed as:

$$V_{i,t} = \sum_{l \in \mathscr{L}} \left(\zeta_l \sum_{i' \in \mathscr{N}, i' \neq i} G_{i',i,l,t} + \omega_l \sum_{i' \in \mathscr{N}, i' \neq i} G_{i,i',l,t} \right) \tag{A.12}$$

where ζ_l and ω_l represent the electricity consumption required to bring the CO_2 from transport to reference conditions and viceversa, respectively. They are calculated based on the specific capture and transport conditions for the different technologies, by using Aspen Plus® (Aspen, 2014) and by considering an isentropic efficiency of the compressor of 65%. The electricity consumption is positive when the pressure is increased and when the temperature is reduced, whereas it is equal to zero when compression and cooling are not required. This implies that no energy recovery via expansion and no heating duty are considered. Note that by choosing the reference conditions as the capture conditions, no capture expenditure enters Eq. (A.12).

A4. Objective function

The objective function of the optimization problem is the total cost of the system, J, as expressed by Eq. (2) of the paper. This includes the capital, C, operation, O and maintenance, M contribution for capture, transport and storage technologies.

Capture technologies. The capital cost is computed by multiplying the size of the capture unit, i.e. the rated amount of CO_2 captured, S, by the unit cost of capture per unit CO_2 annualized over the technology lifetime. For all capture sites $i \in \mathscr{N}_C$ and for all $t \in \{0, \ldots, T\}$, the capital cost of capture technologies, $C^C_{i,t}$, is expressed as:

$$C^C_{i,t} = \alpha_{1,i} S_{i,t} f_t \tag{A.13}$$

where f is a discount factor depending on the year of interest, t, and on the discount rate, r:

$$f_t = \frac{1}{(1+r)^t} \tag{A.14}$$

The operation cost of capture technologies, $O^C_{i,t}$, is given by the cost of electricity consumed by the capture process:

$$O^C_{i,t} = u F_{i,t} f_t \tag{A.15}$$

The maintenance cost of capture technologies, $M^C_{i,t}$, is given as a fraction, ϕ, of their capital cost:

$$M^C_{i,t} = \phi_i C^C_{i,t} f_t \tag{A.16}$$

Transport technologies. The capital cost is computed as a function of the size, i.e. the rated amount of CO_2 transported, R, and of the length of the installed connection, L. For all transport technologies $l \in \mathscr{L}$, for all connections between nodes i and i', with $i, i' \in \mathscr{N}$, and for all $t \in \{0, \ldots, T\}$, the capital cost of transport technologies, $C^L_{i,i',l,t}$, is expressed as:

$$C^L_{i,i',l,t} = \left(\alpha_{1,i,i',l} R_{i,i',l,t} + \alpha_{2,l} L_{i,i',l} y_{i,i',l,t} + \alpha_{0,l} y_{i,i',l,t} \right) f_t \tag{A.17}$$

where the parameter $\alpha_{1,i,i',l}$ depends not only on the transport technology, but also on the specific connection (see Table B.2 in Appendix B).

The operation cost of transport technologies, $O^{\mathrm{L}}_{i,i',l,t}$, is computed as a function of the flow, i.e. the actual amount of CO_2 transported, G, and of the length of the operated connection:

$$O^{\mathrm{L}}_{i,i',l,t} = \left(\beta_{1,l} G_{i,i',l,t} L_{i,i',l} + \beta_{2,l} G_{i,i',l,t}\right) f_t \tag{A.18}$$

The maintenance cost of transport technologies, $M^{\mathrm{L}}_{i,i',l,t}$, is given as a fraction of their capital cost, according to Eq. (A.16) for the transport technologies.

Storage technologies. The capital cost is computed by multiplying the amount of CO_2 stored, U, by the unit cost of the storage facility annualized over its lifetime. For all storage sites $i \in \mathcal{N}_{\mathrm{S}}$ and for all $t \in \{0, \ldots, T\}$, the capital cost of storage technologies, $C^{\mathrm{S}}_{i,t}$, is expressed as:

$$C^{\mathrm{S}}_{i,t} = \alpha_1 U_{i,t} f_t \tag{A.19}$$

The operation cost of storage technologies, $O^{\mathrm{S}}_{i,t}$, is given by the cost of electricity consumed by the storage process, according to Eq. (A.15) for the storage sites. Similarly, the maintenance cost of storage technologies, $M^{\mathrm{S}}_{i,t}$, is given as a fraction of their capital cost, according to Eq. (A.16) for the storage sites.

The total capital, operation and maintenance costs appearing in Eq. (2) are obtained by summing all contributions (including the cost related to the electricity used for conditioning the CO_2) over all nodes and technologies:

$$C_t = \sum_{i \in \mathcal{N}_{\mathrm{C}}} C^{\mathrm{C}}_{i,t} + \sum_{i \in \mathcal{N}_{\mathrm{S}}} C^{\mathrm{S}}_{i,t} + \sum_{i \in \mathcal{N}} \sum_{i' \in \mathcal{N}, i' \neq i} \sum_{l \in \mathcal{L}} C_{i,i',l,t} \tag{A.20}$$

$$O_t = \sum_{i \in \mathcal{N}_{\mathrm{C}}} O^{\mathrm{C}}_{i,t} + \sum_{i \in \mathcal{N}_{\mathrm{S}}} O^{\mathrm{S}}_{i,t} + \sum_{i \in \mathcal{N}} \sum_{i' \in \mathcal{N}, i' \neq i} \sum_{l \in \mathcal{L}} O_{i,i',l,t} + u \sum_{i \in \mathcal{N}} V_{i,t} \tag{A.21}$$

$$M_t = \sum_{i \in \mathcal{N}_{\mathrm{C}}} M^{\mathrm{C}}_{i,t} + \sum_{i \in \mathcal{N}_{\mathrm{S}}} M^{\mathrm{S}}_{i,t} + \sum_{i \in \mathcal{N}} \sum_{i' \in \mathcal{N}, i' \neq i} \sum_{l \in \mathcal{L}} M_{i,i',l,t} \tag{A.22}$$

While minimizing the total cost of the system, the optimization algorithm is constrained by a maximum value of the total CO_2 emissions of the systems (see Sections 2 and 3 of the paper). For the capture and storage sites, CO_2 emissions are due to the electricity consumed to capture and store the CO_2, respectively. For the transport technologies, the CO_2 emissions include both the direct emissions resulting from the network operation and the indirect emissions embodied in the technology installation. The emissions resulting from conditioning the CO_2 are also included. Accordingly, the total CO_2 emissions of the system, which must comply with the LER and CER emissions pathways defined in Section 3 of the paper, are expressed as:

$$e_t = \sum_{i \in \mathcal{N}} \epsilon_i \left(F_{i,t} + V_{i,t}\right) \sum_{i \in \mathcal{N}} \sum_{i' \in \mathcal{N}, i' \neq i} \sum_{l \in \mathcal{L}} L_{i,i',l} \left(\gamma_{1,l} G_{i,i',l,t} + \gamma_{2,l} R_{i,i',l,t}\right) \tag{A.23}$$

where the parameters ϵ_i, $\gamma_{1,l}$, and $\gamma_{2,l}$ quantify the total CO_2 emissions per unit grid electricity in node i, the direct CO_2 emissions per unit length and CO_2 transported by technology l, and the indirect CO_2 emissions per unit length and capacity installed of technology l, respectively.

A5. Reduction of complexity and numerical solver

For n nodes the number of possible connections via a given transport mode is on the order of $\mathcal{O}(n(n-1))$. To limit the computational complexity of the optimization problem, the number of possible connections has been reduced by means of Delaunay triangulation (Morbee et al., 2012), which allows to reduce the number of possible connections to $\mathcal{O}(n)$. All instances of the optimization problem are solved by considering a MIP gap of 1% and by using the commercial solver Gurobi (Gurobi Optimization, 2021).

Appendix B. Numerical values of the input data to the optimization problem

A constant value is considered for the electricity price, $p = 100$ EUR/MWh (Gabrielli et al., 2022), whereas two different values are considered for the electricity carbon intensity, $\epsilon_i = 104$ gCO_2/kWh when node i is in Switzerland, and $\epsilon_i = 414$ gCO_2/kWh when node i is outside Switzerland (Moro and Lonza, 2018; Wernet et al., 2016).

The numerical values of the parameters characterizing the performance of the capture, transport and storage technologies, as well as the conditions at which CO_2 is captured, transported and stored, are reported in Table B.1 (Equinor, 2019; IEAGHG, 2013; Personal Communications with Industry Experts from Neustark, 2021; Zep, 2011). A maximum capture rate of $\psi = 0.9$ is considered for all capture units.

The symbols T, p, and v indicate the temperature, the pressure, and the average travel speed characterizing the CO_2 transport technologies. Note that a value $\eta = 0$ is considered for the storage sites because the electricity consumption required for storing the CO_2 depends on the temperature and pressure of the connection arriving to the storage site, and is accounted for through the coefficients ζ and ω (see Eq. (A.12)).

The numerical values of the parameters characterizing the cost of the capture, transport and storage technologies are reported in Table B.2 (DB Cargo AG, 2021; Personal Communications with Industry Experts from Neustark, 2021; Zep, 2011).

For the transport technology except pipeline, the cost coefficient α_1 is obtained by (i) considering a cost of the isotainer of $c = 4500$ EUR/tCO_2 (Personal Communications with Industry Experts from Neustark, 2021), (ii) actualizing it over the technology lifetime, δ, (iii) accounting for the number of trips per year that a connection is exploited for, and (iv) multiplying the resulting value by the length of the corresponding connection (generically denoted as L in Table B.2). This is expressed as:

Table B.1
Numerical values of the parameters describing the technology performance and the conditions at which CO_2 is captured, transported and stored. Legend: * Minimum of one truck per day, ** Minimum of one barge/ship per week.

Technology	η [kWh_e/tCO_2]	R^{min} [tCO_2/y]	R^{max} [tCO_2/y]	T [K]	p [bar]	λ [1/m]	v [km/h]
WtE CO_2 capture unit (Ahn et al., 2013)	25			313.15	1.5		
CO_2 storage site	0			263.15	150		
Truck (Personal Communications with Industry Experts from Neustark, 2021)		1.6×10^4 *	10^8	238.15	22	10^{-9}	40
Train (Personal Communications with Industry Experts from Neustark, 2021)		1.6×10^4 *	10^8	238.15	22	10^{-9}	19
Barge (Equinor, 2019)		4.2×10^5 **	10^8	238.15	22	10^{-9}	9
Ship (Equinor, 2019)		4.2×10^5 **	10^8	238.15	22	10^{-9}	18
Pipeline (IEAGHG, 2013; Zep, 2011)		2.8×10^5	10^8	298.15	110	2×10^{-9}	

Table B.2
Numerical values of the parameters describing the technology cost.

Technology	α_0 [EUR]	α_1 [EUR/tCO_2]	α_2 [EUR/km]	β_1 [EUR/tCO_2/km]	β_2 [EUR/tCO_2]	ϕ [-]	δ [y]
WtE CO_2 capture unit	50					0.01	20
CO_2 storage site	5					1	40
Truck (Personal Communications with Industry Experts from Neustark, 2021)	100	$0.004L$	0	0.48	44.3	0	10
Train (CH) (DB Cargo AG, 2021)	10	$0.008L$	0	0.16	30.0	0	10
Train (EU) (DB Cargo AG, 2021)	10	$0.008L$	0	0.16	30.0	0	10
Barge (Zep, 2011)	10	$0.017L$	0	0.06	0	0	10
Ship (Zep, 2011)	10	$0.009L$	0	0.06	0	0	10
Pipeline (Zep, 2011)	100	10^{-6}	10^5	0.008	0	0	30

Table B.3
Parameters describing the emissions of network technologies.

Technology	γ_1 [$tCO_2/tCO_2/km$]	γ_2 [tCO_2-Eq/tCO_2/km]
Truck	6.83×10^{-5}	1.50×10^{-5}
Train (CH)	6.64×10^{-7}	1.20×10^{-5}
Train (EU)	1.98×10^{-5}	1.90×10^{-5}
Barge	9.60×10^{-7}	4.40×10^{-5}
Ship	9.60×10^{-7}	8.40×10^{-6}
Pipeline	1.55×10^{-6}	3.15×10^{-6}

Table B.4
CO_2 emissions of all Swiss WtE plants in 2018.

Node	CO_2 emissions [tCO_2/y]
Basel	257,125
Geneva	230,402
Giubiasco	188,190
Trimmis	124,446
Colombier	64,740
Zuchwil	252,071
Weinfelden	163,825
Winterthur	224,196
Buchs (AG)	157,025
Oftringen	80,390
Turgi	134,263
Bern	156,229
Biel	55,004
Thun	146,573
Posieux	114,048
Niederurnen	123,038
La Chaux-de-Fonds	60,129
Buchs (SG)	208,719
St. Gallen	82,044
Lausanne	209,154
Gamsen	43,400
Sion	65,319
Monthey	164,588
Dietikon	105,275
Hinwil	216,715
Horgen	41,910
ZH Hagenholz	270,227
ZH Josefstrasse	122,829
Luzern	302,944
Bazenheid	98,558

$$\alpha_{1,i,i',l} = c\frac{r}{1-\left(\frac{1}{1+r}\right)^{\delta}}\frac{8000v_l}{2L_{i,i',l}} \tag{B.1}$$

where 8000 is the number of hours where the connection is active per year and the factor 2 accounts for the round-trip expenditures (e.g., it includes the fuel consumption for the return trip of the empty carriage). A discount rate of $r = 0.06$ is considered for all technologies.

The numerical values of the parameters characterizing the emissions of the transport technologies are reported in Table B.3 (Cefic and ECTA, 2011; Equinor, 2019; McKinnon and Piecyk, 2011).

The CO_2 emissions of all WtE plants are reported in Table B.4 (Swiss Federal Office of the Environment, 2021).

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